Number Theory for Competitions
A Concept Map and Study Notes

What each theorem means, how it is proved, what it is for,
and how real olympiad problems call for it

Source and scope. The order of topics follows the table of contents of T. Andreescu, G. Dospinescu, O. Mushkarov, *Number Theory: Concepts and Problems* (XYZ Press, 2017), Chapters 2–7. Only the contents pages were available when these notes were written, so every explanation, proof and solution below is written independently and is not taken from the book's text. Competition problems are from the public record of the International Mathematical Olympiad.

> **Thesis.** Olympiad number theory is a small number of ideas used in combination: *reduce modulo something*, *compare sizes*, *compare prime powers*, and *use the cyclic structure of powers*. Every named theorem is a tool for one of these four moves. Learn the tool, then learn the signal in a problem statement that asks for it.

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1 How to use these notes

Prerequisites. School algebra, proof by induction, the binomial theorem, and comfort with the notation $a \mid b$ ("a divides b").

How each theorem is presented. A blue box states the result with its exact hypotheses. Then come four short parts: *what it really says* (the meaning in plain words), a *proof*, *use it when* (the signals in a problem), and *why it matters*. Worked examples follow the tools they use.

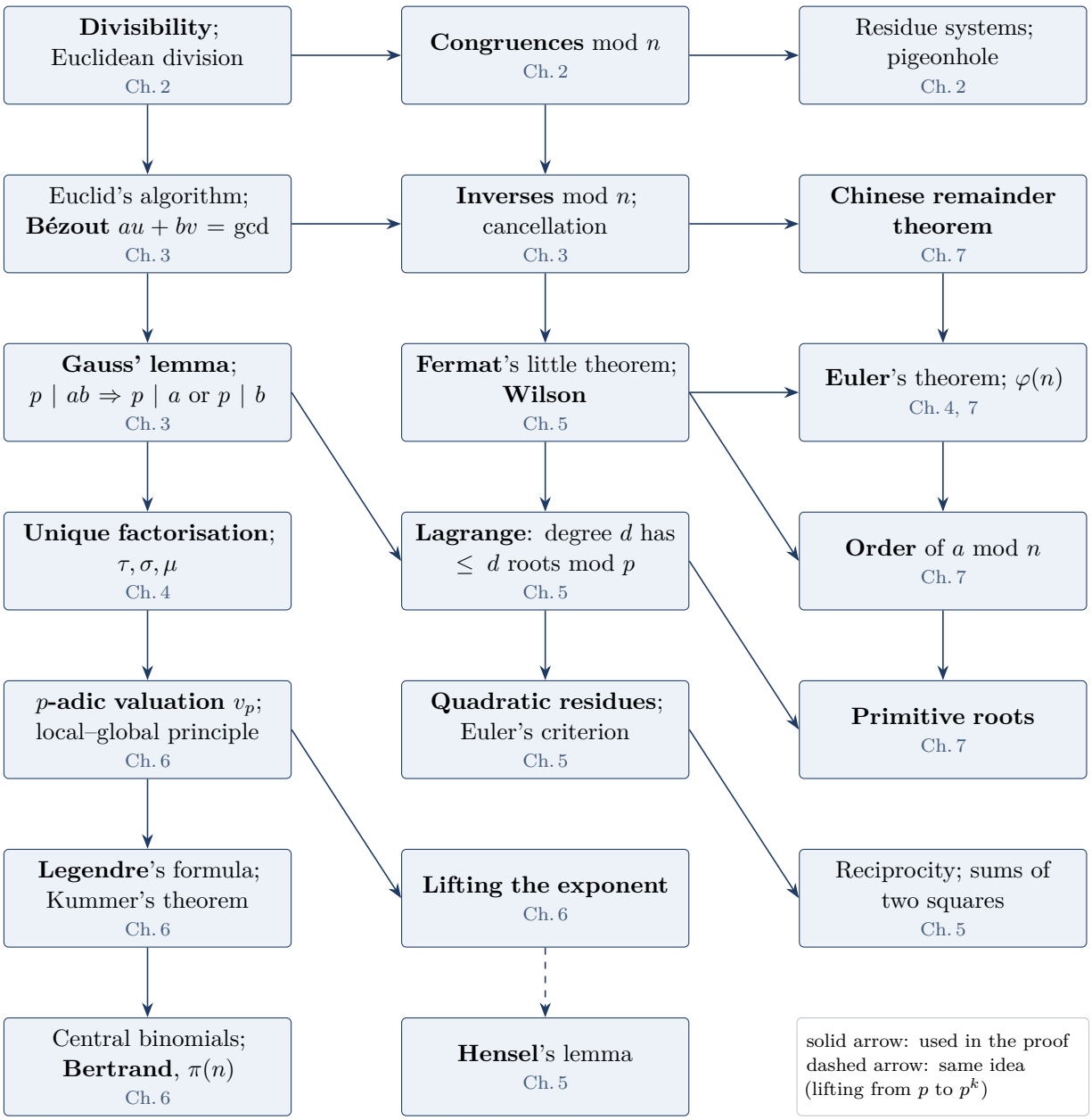
How to study. Read Section 2 first and keep the two maps open. For each theorem, cover the proof and try to reproduce its one key step. In Section 9, read only the problem and the "how to think" box, try for twenty minutes, and only then read the solution.

What is left out. The book's contents list several advanced items that are not treated here: Farey fractions, the Chevalley–Warning theorem, Gauss sums and points on spheres, the congruences of Fleck and Morley, and covering systems. Quadratic reciprocity, Pell's equation and Bertrand's postulate are stated and used, with their proofs omitted or only outlined.

2 The two maps

2.1 Map 1: how the theorems depend on each other

An arrow $A \to B$ means "A is used in the standard proof of B". The left column is about the *multiplicative structure* of integers (factorisation). The middle and right columns are about *congruences*. The chapter tags refer to the book's chapters.

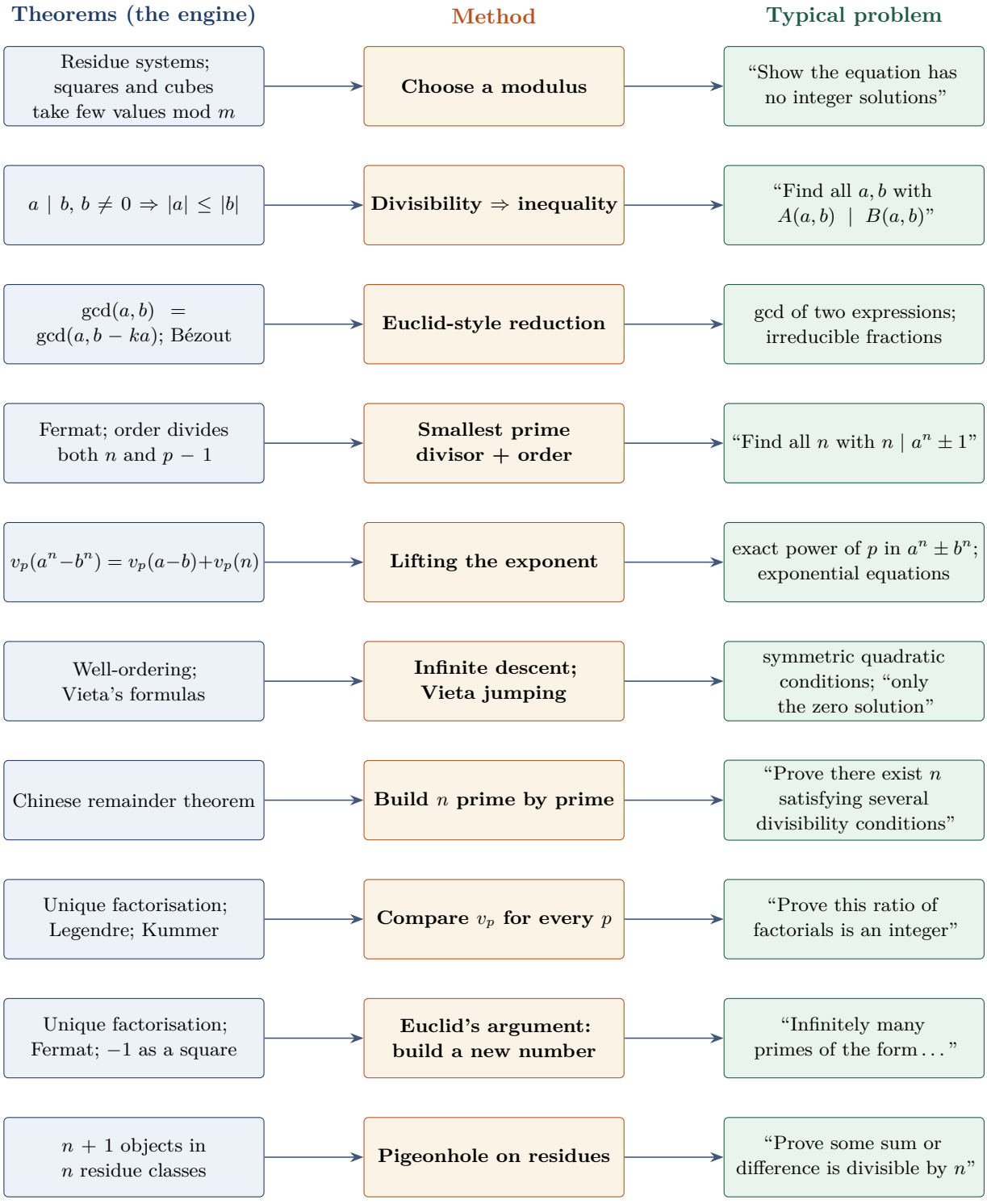


How to read it. Three chains carry almost everything.

- **Left chain.** Division with remainder gives Bézout; Bézout gives Gauss' lemma; Gauss' lemma gives unique factorisation; unique factorisation lets us speak of the exponent $v_p(n)$ of each prime. Legendre's formula and lifting the exponent are computations of v_p.
- **Middle chain.** Bézout also says that a has an inverse modulo n when $\gcd(a, n) = 1$. Inverses give Fermat and Wilson. Because the integers modulo a prime form a field, polynomials behave normally there (Lagrange), which leads to quadratic residues.
- **Right chain.** Inverses give the Chinese remainder theorem, which splits a composite modulus into prime powers. Fermat and Euler say powers are periodic; the exact period is the order; primitive roots say the period can be as long as possible modulo a prime.

2.2 Map 2: methods, the theorems that power them, and the problems they solve

A *method* is a way of attacking a problem; a *theorem* is what makes the method legitimate. Read each row left to right.



2.3 Three questions to ask of every problem

The framework.

1. **What structure is visible?** An exponent that varies, a divisibility between two expressions, a symmetric equation, a factorial, a claim that something exists.
2. **What should I measure?** A residue modulo a well-chosen m; the size of both sides; the power v_p of one prime; the order of a base.
3. **What would finish the proof?** An impossible residue; an inequality that leaves finitely many cases; equal valuations at every prime; a smaller solution that cannot exist; an explicit construction.

Most solutions use two measurements in succession. A typical pattern is: a congruence shows a certain prime must divide n, a valuation count bounds how often, and a size argument finishes.

3 Divisibility and congruences (Chapter 2)

3.1 The two languages

We write $a \mid b$ if $b = ac$ for some integer c, and $a \equiv b \pmod{n}$ if $n \mid a - b$.

What it really says. A congruence is an equality in which multiples of n have been forgotten. There are only n different "numbers" modulo n, so a statement about all integers becomes a finite check.

Basic rules

1. If $d \mid a$ and $d \mid b$, then $d \mid ax + by$ for all integers x, y.
2. If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.
3. If $a \equiv b$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d$, $ac \equiv bd$, and $a^k \equiv b^k \pmod{n}$.
4. For integers a, b and a polynomial P with integer coefficients, $a - b \mid P(a) - P(b)$. In particular $a - b \mid a^n - b^n$, and $a + b \mid a^n + b^n$ when n is odd.

Proof. (1) and (3) are direct from the definitions; for the product, $ac - bd = (a - b)c + b(c - d)$. (2) $b = ac$ with $c \neq 0$, so $|b| = |a||c| \geq |a|$. (4) $a^k - b^k = (a - b)(a^{k-1} + a^{k-2}b + \cdots + b^{k-1})$; add these over the terms of P. Replace b by $-b$ for the last claim. □

Use it when. Rule 1 removes terms: from $d \mid A$ and $d \mid B$ pass to a simpler combination. Rule 2 is the bridge from divisibility to inequalities and is the standard way to cut a "find all" problem to finitely many cases. Rule 4 is the first thing to try when polynomials with integer coefficients appear.

Division is not free. From $ac \equiv bc \pmod{n}$ you may cancel c only when $\gcd(c, n) = 1$ (Section 4.2). For example $2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}$ but $1 \not\equiv 4 \pmod{6}$.

3.2 Euclidean division and residues of powers

Euclidean division

For integers a and $b > 0$ there are unique integers q, r with $a = bq + r$ and $0 \le r < b$.

What it really says. Every integer falls into exactly one of b classes $0, 1, \ldots, b-1$. This is why case analysis modulo b is a complete proof.

Proof. Take $q = \lfloor a/b \rfloor$, so $bq \le a < bq + b$. If $bq + r = bq' + r'$ then $b \mid r - r'$ with $|r - r'| < b$, so $r = r'$ by Rule 2. □

Powers take few values for suitable moduli. These short lists settle a large share of "no solutions" problems and should be known by heart.

Expression	Modulus	Possible residues
x^2	3, 4	$0, 1$
x^2	8	$0, 1, 4$ (odd squares are $\equiv 1$)
x^2	16	$0, 1, 4, 9$
x^3	7, 9	$0, \pm 1$
x^4	16	$0, 1$

Example 3.1. *Prove that $x^2 + y^2 = 4z + 3$ has no integer solutions.*

Proof. Modulo 4 a square is 0 or 1, so $x^2 + y^2 \in \{0, 1, 2\}$, never 3. □

Choosing the modulus. Pick m so that the varying part takes few values: 4 or 8 for squares, 7 or 9 for cubes, 16 for fourth powers, and for k-th powers a prime p with $k \mid p - 1$ (then k-th powers take only $(p-1)/k + 1$ values modulo p; see Section 8.4).

3.3 Binomial coefficients

Two facts about binomial coefficients

1. The product of any k consecutive integers is divisible by $k!$.
2. If p is prime and $0 < k < p$, then $p \mid \binom{p}{k}$. Hence $(a + b)^p \equiv a^p + b^p \pmod{p}$.

Proof. (1) For positive integers, $n(n-1)\cdots(n-k+1)/k! = \binom{n}{k}$ counts subsets, so it is an integer; the cases with zero or negative factors reduce to this one. (2) From $k\binom{p}{k} = p\binom{p-1}{k-1}$ we get $p \mid k\binom{p}{k}$; since p is prime and $p \nmid k$, $p \mid \binom{p}{k}$ (Euclid's lemma, Section 4.2). □

Why it matters. Fact 2 is the reason raising to the p-th power is additive modulo p. Fermat's theorem, Lucas' theorem, lifting the exponent and Hensel's lemma all start from expanding $(a+b)^p$ or $(a + tp^k)^n$ and discarding terms that are visibly divisible by a high power of p.

3.4 Pigeonhole on residues

Example 3.2. *Prove that among any n integers $a_1, \ldots, a_n$ there is a non-empty block $a_{i+1} + \cdots + a_j$ whose sum is divisible by n.*

Proof. Consider the $n+1$ partial sums $s_0 = 0$, $s_k = a_1 + \cdots + a_k$. There are only n residues modulo n, so $s_i \equiv s_j$ for some $i < j$, and then $n \mid s_j - s_i$. □

Transferable lesson. When asked to prove that *some* combination is divisible by n, create more than n candidates and subtract two that agree modulo n.

4 Greatest common divisors (Chapter 3)

4.1 Bézout's theorem and Euclid's algorithm

Bézout's theorem

For integers a, b not both zero, the set $\{ax + by : x, y \in \mathbb{Z}\}$ is exactly the set of multiples of $d = \gcd(a, b)$. In particular there are integers u, v with $au + bv = d$.

What it really says. The gcd is not only the largest common divisor; it is the smallest positive number that can be *built* from a and b by integer combinations. The equation $au + bv = 1$ is a certificate that a and b are coprime, and unlike the definition of gcd it is an equation you can multiply, substitute and reduce.

Proof. Let $S = \{ax + by\}$ and let d be its smallest positive element. Divide a by d: $a = dq + r$, $0 \le r < d$. Then $r = a - dq \in S$, so $r = 0$ by minimality, and $d \mid a$. Likewise $d \mid b$. Any common divisor of a and b divides every element of S, in particular d, so $d = \gcd(a, b)$. Every element of S is a multiple of d, and every multiple of d is in S. □

Euclid's algorithm. Since $\gcd(a, b) = \gcd(b, a - qb)$, replacing the larger number by a remainder keeps the gcd and shrinks the numbers. The same step works for expressions: $\gcd(A, B) = \gcd(A - kB, B)$ for any integer k.

Use it when. A problem says "relatively prime", "irreducible fraction", or asks for the gcd of two expressions in n. Subtract multiples to lower the degree until a constant remains.

Example 4.1. *Find* $\gcd(2^m - 1,\ 2^n - 1)$.

Proof. If $m = qn + r$ then $2^m - 1 = 2^r(2^{qn} - 1) + (2^r - 1)$ and $2^n - 1 \mid 2^{qn} - 1$, so $\gcd(2^m - 1, 2^n - 1) = \gcd(2^n - 1, 2^r - 1)$. The exponents follow Euclid's algorithm on (m, n), so the answer is $2^{\gcd(m,n)} - 1$. □

4.2 Gauss' lemma and inverses

Gauss' lemma, Euclid's lemma, and inverses

1. (Gauss) If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.
2. (Euclid) If a prime p divides ab, then $p \mid a$ or $p \mid b$.
3. If $\gcd(a, b) = 1$, $a \mid n$ and $b \mid n$, then $ab \mid n$.
4. (Inverse) a has an inverse modulo n, i.e. some x with $ax \equiv 1 \pmod{n}$, if and only if $\gcd(a, n) = 1$.

Proof. (1) Take $au + bv = 1$ and multiply by c: $c = acu + bcv$, and a divides both terms. (2) If $p \nmid a$ then $\gcd(p, a) = 1$; apply (1). (3) Write $n = ak$; then $b \mid ak$ and $\gcd(a, b) = 1$ give $b \mid k$. (4) $ax \equiv 1 \pmod{n}$ means $ax + ny = 1$ for some y, which by Bézout happens exactly when $\gcd(a, n) = 1$. □

What it really says. Coprime factors do not interfere: a divisor that shares nothing with b must take everything it needs from c. Part 4 says that modulo n you may divide by exactly the numbers

coprime to n. Modulo a prime you may divide by anything non-zero, so arithmetic modulo p behaves like arithmetic with fractions.

Consequence (permutation). If $\gcd(a, n) = 1$, then $x \mapsto ax$ permutes the residues modulo n, because $ax \equiv ay$ forces $x \equiv y$. This one observation proves Fermat's and Euler's theorems.

4.3 Linear equations, Pythagorean triples, rational roots

Linear Diophantine equation

$ax + by = c$ has integer solutions if and only if $d = \gcd(a, b)$ divides c. If (x_0, y_0) is one solution, all solutions are $x = x_0 + \frac{b}{d}t$, $y = y_0 - \frac{a}{d}t$, $t \in \mathbb{Z}$.

Proof. Solvability is Bézout. If (x, y) is another solution, $a(x - x_0) = -b(y - y_0)$; divide by d and apply Gauss' lemma to get $\frac{b}{d} \mid x - x_0$. □

Primitive Pythagorean triples

The positive solutions of $x^2 + y^2 = z^2$ with $\gcd(x, y) = 1$ and y even are exactly $x = m^2 - n^2$, $y = 2mn$, $z = m^2 + n^2$ with $m > n > 0$ coprime and of opposite parity.

Proof. Here x, z are odd, and $\left(\frac{y}{2}\right)^2 = \frac{z-x}{2} \cdot \frac{z+x}{2}$. The two factors are coprime (a common divisor divides their sum z and difference x). Two coprime positive integers whose product is a square are both squares (Section 5.1), say n^2 and m^2. Solving gives the formulas; the converse is a direct check. □

Transferable lesson. Factor, show the factors are coprime, and conclude each factor is itself a perfect power. This is the most common way to solve equations of the form (product) = (perfect power).

Rational root theorem

If $a_n x^n + \cdots + a_0$ has integer coefficients and a rational root $\frac{p}{q}$ in lowest terms, then $p \mid a_0$ and $q \mid a_n$. In particular a rational root of a monic integer polynomial is an integer.

Proof. Multiply $P(p/q) = 0$ by q^n: $a_n p^n + a_{n-1} p^{n-1} q + \cdots + a_0 q^n = 0$. All terms but the first contain q, so $q \mid a_n p^n$, and $\gcd(q, p^n) = 1$ gives $q \mid a_n$. Symmetrically $p \mid a_0$. □

Why it matters. It reduces "is this number rational?" to a finite check. For example $\sqrt[k]{n}$ is a root of $x^k - n$, so it is either an integer or irrational.

4.4 Least common multiple and Pell's equation

lcm. $\operatorname{lcm}(a, b)$ is the positive integer m characterised by: $a \mid n$ and $b \mid n$ if and only if $m \mid n$. For positive a, b, $\gcd(a, b) \cdot \operatorname{lcm}(a, b) = ab$. Both statements are clearest prime by prime: gcd takes the minimum exponent, lcm the maximum (Section 5.1).

Pell's equation (stated; existence not proved here)

If $d > 0$ is not a perfect square, $x^2 - dy^2 = 1$ has infinitely many positive integer solutions. If (x_1, y_1) is the smallest, all are given by $x_k + y_k\sqrt{d} = (x_1 + y_1\sqrt{d})^k$.

What it really says. Solutions multiply. The identity

$$(x_1^2 - dy_1^2)(x_2^2 - dy_2^2) = (x_1x_2 + dy_1y_2)^2 - d(x_1y_2 + x_2y_1)^2$$

shows that two solutions combine into a third, so one non-trivial solution produces infinitely many. The existence of a first solution is the hard part and uses a pigeonhole argument on approximations of $\sqrt{d}$.

Use it when. A problem asks for infinitely many integers satisfying a quadratic relation, for example "infinitely many n such that n and $n+1$ are both ..." or triangular numbers that are squares. Reduce to $x^2 - dy^2 = 1$ or $= c$ and quote one small solution.

4.5 Infinite descent and Vieta jumping

The move. Assume a solution exists and take one that is smallest in some positive-integer measure. Construct from it a strictly smaller solution of the same kind. Since positive integers cannot decrease for ever, no solution exists (or only the trivial one, or the solution must be of a special form on which the construction breaks).

Example 4.2. *Prove that $x^3 = 2y^3 + 4z^3$ has only the integer solution $x = y = z = 0$.*

Proof. Suppose there is a non-zero solution, and choose one with $|x| + |y| + |z|$ minimal. The right side is even, so $x = 2x_1$. Then $8x_1^3 = 2y^3 + 4z^3$, so $y^3 = 4x_1^3 - 2z^3$ is even and $y = 2y_1$. Then $8y_1^3 = 4x_1^3 - 2z^3$, so $z^3 = 2x_1^3 - 4y_1^3$ is even and $z = 2z_1$. Substituting, $8z_1^3 = 2x_1^3 - 4y_1^3$, that is $x_1^3 = 2y_1^3 + 4z_1^3$. So (x_1, y_1, z_1) is a non-zero solution with smaller measure, a contradiction. □

Vieta jumping is descent for conditions that are quadratic in each variable. Fix all variables but one; the condition becomes a quadratic $x^2 - sx + t = 0$ with a known integer root a. By Vieta's formulas the other root is $a' = s - a = t/a$: the first expression shows a' is an integer, the second lets you estimate its size. If a' is positive and smaller than a, you have descended. The standard example is Problem 5 in Section 9.

What a descent must verify. The new object must be (i) an integer, (ii) admissible (positive, non-zero, whatever the problem requires), and (iii) strictly smaller. Each of the three needs its own sentence in a written proof.

5 The fundamental theorem of arithmetic (Chapter 4)

5.1 Unique factorisation

Fundamental theorem of arithmetic

Every integer $n > 1$ is a product of primes, and the product is unique up to the order of the factors.

What it really says. Under multiplication, a positive integer *is* its list of prime exponents: $n = \prod p^{v_p(n)}$. Multiplying adds the lists, and

$$a \mid b \iff v_p(a) \le v_p(b) \text{ for every prime } p, \qquad \gcd \leftrightarrow \min, \qquad \operatorname{lcm} \leftrightarrow \max .$$

A question about divisibility becomes a separate, usually easier, question for each prime.

Proof. Existence. By strong induction: n is prime, or $n = ab$ with $1 < a, b < n$ and both factor. *Uniqueness.* If $p_1 \cdots p_r = q_1 \cdots q_s$, then p_1 divides the right side, so by Euclid's lemma p_1 divides some q_j, hence equals it. Cancel and repeat. □

First consequences

1. If $\gcd(a, b) = 1$ and ab is a perfect k-th power (with $a, b > 0$), then a and b are perfect k-th powers.
2. If $n = p_1^{a_1} \cdots p_r^{a_r}$, the number of positive divisors is $\tau(n) = (a_1 + 1) \cdots (a_r + 1)$ and their sum is $\sigma(n) = \prod_i \frac{p_i^{a_i+1}-1}{p_i-1}$.
3. $\tau(n)$ is odd if and only if n is a perfect square.
4. A composite n has a prime divisor $p \le \sqrt{n}$.

Proof. (1) Each prime divides only one of a, b, and its exponent in ab is a multiple of k. (2) A divisor is a choice of exponent $0 \le e_i \le a_i$ for each prime; expanding $\prod(1 + p_i + \cdots + p_i^{a_i})$ lists every divisor once. (3) The product in (2) is odd exactly when every a_i is even. Alternatively pair d with n/d. (4) If $n = ab$ with $1 < a \le b$ then $a \le \sqrt{n}$; take a prime factor of a. □

5.2 The smallest prime divisor

The move. When a condition involves all of n (for example $n \mid 2^n - 1$), look only at the smallest prime p dividing n. Its advantage: every prime factor of $p - 1$ is smaller than p, so $\gcd(n, p - 1) = 1$.

This is used with the notion of order in Section 8.3; see Example 8.1 and Problems 7 and 8.

A frequent false step. What is proved for the smallest prime divisor is not automatically true for the others. After finding it, divide it out and argue again for the next smallest.

5.3 Infinitely many primes: Euclid's argument

Euclid

There are infinitely many primes.

Proof. Given primes $p_1, \ldots, p_k$, the number $N = p_1 \cdots p_k + 1$ exceeds 1, so it has a prime factor, and that factor is none of the p_i because each leaves remainder 1. □

What it really says. To produce a new prime with a wanted property, build a number N that is coprime to all known primes and whose prime factors are *forced* to have the property.

Example 5.1. *There are infinitely many primes of the form $4k + 3$, and infinitely many of the form $4k + 1$.*

Proof. Form $4k + 3$. Let $N = 4p_1 \cdots p_k - 1$. It is odd and $\equiv 3 \pmod 4$. A product of primes that are all $\equiv 1 \pmod 4$ is $\equiv 1$, so N has a prime factor $\equiv 3 \pmod 4$, different from every p_i.

Form $4k + 1$. Let $N = (2p_1 \cdots p_k)^2 + 1$. Any prime $q \mid N$ is odd and satisfies $x^2 \equiv -1 \pmod q$,

which forces $q \equiv 1 \pmod 4$ (Theorem in Section 6.4). Again q is none of the p_i. □

5.4 Arithmetic functions

A function f on positive integers is *multiplicative* if $f(mn) = f(m)f(n)$ whenever $\gcd(m,n) = 1$. Then f is determined by its values on prime powers. The functions τ, σ, φ, μ are multiplicative.

Euler's function and the Möbius function

1. $\varphi(n)$, the number of $1 \le k \le n$ with $\gcd(k,n) = 1$, equals $n\prod_{p\mid n}\left(1-\frac{1}{p}\right)$.
2. $\sum_{d\mid n}\varphi(d) = n$.
3. $\mu(n) = (-1)^r$ if n is a product of r distinct primes, and 0 if n is divisible by a square > 1. Then $\sum_{d\mid n}\mu(d) = 1$ if $n = 1$ and 0 otherwise.

Proof. (1) $\varphi(p^a) = p^a - p^{a-1}$, since only the multiples of p are excluded; multiplicativity follows from the Chinese remainder theorem (Section 8.1). (2) Write the fractions $\frac{1}{n}, \frac{2}{n}, \ldots, \frac{n}{n}$ in lowest terms. The denominators are the divisors d of n, and denominator d occurs exactly $\varphi(d)$ times. (3) If n has $r \ge 1$ distinct primes, only squarefree divisors count, and the sum is $\sum_j \binom{r}{j}(-1)^j = (1-1)^r = 0$.□

Why it matters. φ is the exponent in Euler's theorem and counts invertible residues. Identity (2) is the counting step in the proof that primitive roots exist. μ encodes inclusion–exclusion over prime divisors: it is the tool for counting integers coprime to n, or squarefree integers, in a range.

6 Congruences modulo a prime (Chapter 5)

Throughout this section p is a prime.

6.1 Fermat's little theorem

Fermat's little theorem

If $p \nmid a$ then $a^{p-1} \equiv 1 \pmod p$. For every integer a, $a^p \equiv a \pmod p$.

What it really says. Powers modulo p are periodic, with period dividing $p-1$. So in $a^n \bmod p$ only $n \bmod (p-1)$ matters: bases live modulo p, exponents live modulo $p-1$.

Proof. Multiplication by a permutes the non-zero residues $1, 2, \ldots, p-1$. So $a \cdot 2a \cdots (p-1)a \equiv 1 \cdot 2 \cdots (p-1)$, that is $a^{p-1}(p-1)! \equiv (p-1)!$. Cancel $(p-1)!$, which is coprime to p. □

Use it when. (i) A huge exponent must be reduced. (ii) You must show a given prime divides some term of a sequence built from powers: choose the index in terms of p, typically $n = p-1$ or $n = p-2$, so that every power collapses (Problem 9). (iii) As the first step toward the order of an element.

> **Hypothesis.** $a^{p-1} \equiv 1$ needs $p \nmid a$. The converse is false: $a^{n-1} \equiv 1 \pmod n$ can hold for composite n (for instance $2^{340} \equiv 1 \pmod{341}$, $341 = 11 \cdot 31$).

6.2 Wilson's theorem

Wilson's theorem

An integer $n > 1$ is prime if and only if $(n-1)! \equiv -1 \pmod{n}$.

Proof. Let $n = p$ be prime. Pair each $x \in \{1, \dots, p-1\}$ with its inverse. The only self-paired elements satisfy $x^2 \equiv 1$, i.e. $p \mid (x-1)(x+1)$, so $x \equiv \pm 1$. All other pairs multiply to 1, hence $(p-1)! \equiv 1 \cdot (-1)$. Conversely, if n has a divisor $1 < d < n$, then $d \mid (n-1)!$, so $(n-1)! \equiv -1 \pmod{n}$ would give $d \mid 1$. □

What it really says. It evaluates a product over *all* non-zero residues. Whenever a product of many consecutive integers appears modulo p, try to complete it to $(p-1)!$.

Corollary: a square root of -1

If $p \equiv 1 \pmod{4}$ then $x = \left(\frac{p-1}{2}\right)!$ satisfies $x^2 \equiv -1 \pmod{p}$.

Proof. $(p-1)! = \prod_{k=1}^{(p-1)/2} k\,(p-k) \equiv (-1)^{(p-1)/2} x^2 = x^2$, and the left side is -1. □

6.3 Lagrange's theorem on polynomial congruences

Lagrange

A polynomial of degree d with integer coefficients, not all divisible by p, has at most d roots modulo p. Consequently, if $d \mid p-1$ then $x^d \equiv 1 \pmod{p}$ has exactly d solutions.

What it really says. Modulo a prime, polynomials behave as they do over the real numbers. This fails for composite moduli: $x^2 \equiv 1 \pmod{8}$ has the four roots $1, 3, 5, 7$.

Proof. Induction on d. If r is a root, $f(x) - f(r) = (x-r)g(x)$ with $\deg g = d-1$, so $f(x) \equiv (x-r)g(x)$. Another root $s \not\equiv r$ gives $p \mid (s-r)g(s)$, hence $p \mid g(s)$ by Euclid's lemma; g has at most $d-1$ roots. For the consequence, write $x^{p-1} - 1 = (x^d - 1)h(x)$ with $\deg h = p-1-d$. By Fermat the left side has $p-1$ roots; h has at most $p-1-d$, so $x^d - 1$ has at least d, hence exactly d. □

Use it when. You need to count solutions modulo p, or to show that two polynomials with too many common values modulo p are identical modulo p. For example, $x^{p-1} - 1 \equiv (x-1)(x-2)\cdots(x-p+1) \pmod{p}$ as polynomials, because the difference has degree $< p-1$ and $p-1$ roots; comparing coefficients gives Wilson's theorem and more.

6.4 When is -1 a square? Sums of two squares

-1 as a square

For an odd prime p, the congruence $x^2 \equiv -1 \pmod{p}$ is solvable if and only if $p \equiv 1 \pmod{4}$. Hence if $p \equiv 3 \pmod{4}$ and $p \mid a^2 + b^2$, then $p \mid a$ and $p \mid b$.

Proof. If $x^2 \equiv -1$, then $1 \equiv x^{p-1} = (x^2)^{(p-1)/2} \equiv (-1)^{(p-1)/2}$, so $\frac{p-1}{2}$ is even. The converse is the corollary to Wilson. For the last claim, if $p \nmid b$, multiply $a^2 \equiv -b^2$ by the square of the inverse of b to get a square root of -1. □

Why it matters. This one fact decides a large class of problems. Every odd prime factor of $n^2 + 1$

is $\equiv 1 \pmod 4$; a prime $\equiv 3 \pmod 4$ divides a sum of two squares only to an even power; and an equation such as $y^2 = x^3 + 7$ is attacked by rewriting it as $y^2 + 1 = (x+2)(x^2 - 2x + 4)$ and looking for a factor $\equiv 3 \pmod 4$.

Fermat's two-squares theorem

Every prime $p \equiv 1 \pmod 4$ is a sum of two squares.

Proof. Pick x with $x^2 \equiv -1 \pmod p$ and let $m = \lfloor \sqrt{p} \rfloor$. There are $(m+1)^2 > p$ pairs (a, b) with $0 \le a, b \le m$, so two different pairs give $a + xb \equiv a' + xb' \pmod p$. Put $u = a - a'$, $v = b - b'$: not both zero, $|u|, |v| < \sqrt{p}$, and $u \equiv -xv$. Then $u^2 \equiv x^2 v^2 \equiv -v^2$, so $p \mid u^2 + v^2$ with $0 < u^2 + v^2 < 2p$. Hence $u^2 + v^2 = p$. □

6.5 Quadratic residues

For p odd and $p \nmid a$, the *Legendre symbol* $\left(\frac{a}{p}\right)$ is 1 if $x^2 \equiv a \pmod p$ is solvable (a is a *quadratic residue*) and -1 otherwise; $\left(\frac{a}{p}\right) = 0$ if $p \mid a$.

Euler's criterion and its consequences

1. Exactly half of the non-zero residues, $\frac{p-1}{2}$ of them, are quadratic residues.
2. $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod p$.
3. $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$.
4. $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$ and $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$; so 2 is a square mod p iff $p \equiv \pm 1 \pmod 8$.
5. (Quadratic reciprocity, stated without proof.) For distinct odd primes p, q: $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$.

Proof. (1) $x^2 \equiv y^2$ iff $x \equiv \pm y$, so the $p-1$ non-zero residues give $\frac{p-1}{2}$ distinct squares. (2) Put $t = a^{(p-1)/2}$; then $t^2 \equiv 1$ so $t \equiv \pm 1$. If $a \equiv x^2$ then $t \equiv x^{p-1} \equiv 1$. So the $\frac{p-1}{2}$ residues are roots of $x^{(p-1)/2} - 1$, which by Lagrange has no further roots; non-residues therefore give $t \equiv -1$. (3) follows from (2). The first formula in (4) is (2) with $a = -1$. The formula for $\left(\frac{2}{p}\right)$ and statement (5) need a further argument (Gauss' lemma on least residues, or Gauss sums), which is omitted here. □

What it really says. Being a square modulo p is a multiplicative ± 1 label. Rule (3) says residue $\times$ residue and non-residue $\times$ non-residue are residues, residue $\times$ non-residue is not. Reciprocity turns the question "is p a square modulo q?" into the reversed question, which lets you reduce the numbers as in Euclid's algorithm.

Use it when. A prime divides a value of a quadratic form: $p \mid x^2 - a$, $p \mid x^2 + ay^2$, or after completing the square in $ax^2 + bx + c$. Conclusion: p lies in certain residue classes modulo $4a$.

6.6 Binomial coefficients and fractions modulo p

Lucas' theorem

If $n = \sum n_i p^i$ and $k = \sum k_i p^i$ are written in base p, then $\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p}$.

Proof. Since $(1+x)^p \equiv 1 + x^p \pmod{p}$, repeated use gives $(1+x)^{p^i} \equiv 1 + x^{p^i}$. Hence $(1+x)^n \equiv \prod_i (1+x^{p^i})^{n_i}$. The coefficient of x^k on the right is found by choosing $x^{k_i p^i}$ from the i-th factor, in $\binom{n_i}{k_i}$ ways, and base-p representation is unique. □

Use it when. Parity or divisibility by p of binomial coefficients: $\binom{n}{k}$ is odd iff every binary digit of k is $\leq$ the corresponding digit of n.

Fractions modulo p. If $p \nmid b$, the fraction $\frac{a}{b}$ is read modulo p^k as $a \cdot b^{-1}$. A rational number with denominator coprime to p is $\equiv 0 \pmod{p^k}$ exactly when p^k divides its numerator. As k runs through $1, \ldots, p-1$, so does k^{-1}, which is how sums of reciprocals are evaluated.

Wolstenholme's theorem

For a prime $p \geq 5$, the numerator of $1 + \frac{1}{2} + \cdots + \frac{1}{p-1}$ is divisible by p^2.

Proof. Let H be the sum. Pairing k with $p-k$, $2H = \sum_{k=1}^{p-1} \left(\frac{1}{k} + \frac{1}{p-k}\right) = p \sum_{k=1}^{p-1} \frac{1}{k(p-k)}$. Modulo p, $\frac{1}{k(p-k)} \equiv -\frac{1}{k^2}$, and since inverses permute the residues, $\sum \frac{1}{k^2} \equiv \sum k^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0 \pmod{p}$ for $p \geq 5$. So $2H \equiv 0 \pmod{p^2}$. □

Transferable lesson. Pair k with $p-k$ to extract a factor p; then replace reciprocals by a permutation of the residues. Problem 3 is the same idea.

6.7 Hensel's lemma

Hensel's lemma

Let f have integer coefficients. If $f(a) \equiv 0 \pmod{p}$ and $f'(a) \not\equiv 0 \pmod{p}$, then for every $k \geq 1$ there is exactly one residue x modulo p^k such that $f(x) \equiv 0 \pmod{p^k}$ and $x \equiv a \pmod{p}$.

What it really says. A simple root modulo p can be refined digit by digit to a root modulo $p^2, p^3, \ldots$, as in Newton's method. So for such equations, solvability modulo prime powers is decided modulo the prime.

Proof. Suppose $f(x_k) \equiv 0 \pmod{p^k}$. By the binomial theorem, $f(x_k + tp^k) \equiv f(x_k) + tp^k f'(x_k) \pmod{p^{k+1}}$. Writing $f(x_k) = p^k c$, we need $c + tf'(x_k) \equiv 0 \pmod{p}$, which has exactly one solution t modulo p because $f'(x_k) \equiv f'(a) \not\equiv 0$. □

Use it when. You need a solution modulo p^k (often to feed into the Chinese remainder theorem). Example: if a is a non-zero square modulo an odd prime p, it is a square modulo every p^k, since $f = x^2 - a$ has $f'(x) = 2x \not\equiv 0$.

7 p-adic valuations (Chapter 6)

7.1 The valuation and the local–global principle

For a prime p and a non-zero integer n, $v_p(n)$ is the exponent of p in n: $p^{v_p(n)} \mid n$ but $p^{v_p(n)+1} \nmid n$. For a fraction, $v_p(a/b) = v_p(a) - v_p(b)$.

Rules for v_p

1. $v_p(ab) = v_p(a) + v_p(b)$.
2. $v_p(a+b) \geq \min(v_p(a), v_p(b))$, with *equality* if $v_p(a) \neq v_p(b)$.
3. (Local–global.) $a \mid b$ iff $v_p(a) \leq v_p(b)$ for all primes p. A positive rational is an integer iff all its valuations are ≥ 0, and a perfect k-th power iff all are multiples of k.

Proof. (1) and (3) restate unique factorisation. (2) If $v_p(a) = s < t = v_p(b)$, write $a = p^s a'$, $b = p^t b'$ with $p \nmid a'$: then $a + b = p^s(a' + p^{t-s}b')$ and the bracket is not divisible by p. If $s = t$ the sum is still divisible by p^s. □

What it really says. v_p turns multiplication into addition, like a logarithm that sees only one prime. Rule 2 says that in a sum, the term with the *smallest* valuation wins if it is alone. Rule 3 says a global statement about integers can be proved one prime at a time.

Example 7.1. *For $n > 1$, the number $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ is not an integer.*

Proof. Let $2^k \leq n < 2^{k+1}$. The term $\frac{1}{2^k}$ has $v_2 = -k$, and every other term $\frac{1}{m}$ has $v_2 > -k$, because the next multiple of 2^k is $2^{k+1} > n$. By Rule 2 (the minimum is attained once), $v_2(H_n) = -k < 0$. □

Use it when. You must prove that a quotient is an integer, that a sum of fractions is not, or that a number is or is not a perfect power; or when a single prime clearly controls the problem.

7.2 Lifting the exponent (LTE)

Lifting the exponent lemma

Let a, b be integers and $n \geq 1$.

1. Let p be an **odd** prime with $p \mid a - b$ and $p \nmid ab$. Then
$$v_p(a^n - b^n) = v_p(a - b) + v_p(n).$$
2. Under the same assumptions with $p \mid a+b$ instead, and n **odd**: $v_p(a^n+b^n) = v_p(a+b)+v_p(n)$.
3. For $p = 2$, with a, b odd and n **even**: $v_2(a^n - b^n) = v_2(a - b) + v_2(a + b) + v_2(n) - 1$. (For n odd, $v_2(a^n - b^n) = v_2(a - b)$.)

What it really says. Once p divides $a - b$, raising to the n-th power gains extra factors of p only from the factors of p in n itself, one for one. The power of p in $a^n - b^n$ grows like $\log n$, while $a^n - b^n$ grows exponentially. That mismatch is what makes LTE decisive in problems such as $n^2 \mid 2^n + 1$.

Proof. Step 1: $p \nmid n$. $\frac{a^n-b^n}{a-b} = \sum_{i=0}^{n-1} a^{n-1-i}b^i \equiv na^{n-1} \pmod{p}$ since $a \equiv b$. This is not divisible by p, so $v_p(a^n - b^n) = v_p(a - b)$.

Step 2: $n = p$. Write $a = b + pc$. By the binomial theorem $a^i \equiv b^i + ib^{i-1}pc \pmod{p^2}$, so

$$\sum_{i=0}^{p-1} a^i b^{p-1-i} \equiv pb^{p-1} + pc\, b^{p-2} \cdot \frac{p(p-1)}{2} \equiv pb^{p-1} \pmod{p^2},$$

because p is odd, so $\frac{p(p-1)}{2}$ is a multiple of p. The sum has valuation exactly 1, hence $v_p(a^p - b^p) = v_p(a-b) + 1$.

Step 3: general $n = p^k m$, $p \nmid m$. Apply Step 2 k times (each time to the pair a^{p^j}, b^{p^j}, which still satisfies the hypotheses) and then Step 1. Part 2 follows by replacing b with $-b$.

Part 3. Write $n = 2^k m$ with m odd, $k \geq 1$:

$$a^n - b^n = (a^m - b^m)(a^m + b^m) \prod_{j=1}^{k-1} \left(a^{2^j m} + b^{2^j m}\right).$$

As in Step 1, $v_2(a^m - b^m) = v_2(a-b)$ and $v_2(a^m + b^m) = v_2(a+b)$, because the quotient is a sum of an odd number of odd terms. Each remaining factor is a sum of two odd squares, $\equiv 2 \pmod 4$, and contributes exactly 1. The total is $v_2(a-b) + v_2(a+b) + k - 1$. □

> **Check the hypotheses every time.** (i) p must divide $a - b$ (or $a + b$); if not, first raise to the order to make it so. (ii) $p \nmid a, b$. (iii) $p = 2$ has its own formula. (iv) The "+" version needs n odd.

Example 7.2. *Find $v_3(2^n + 1)$ for odd n, and $v_2(3^n - 1)$.*

Proof. $3 \mid 2 + 1$, so for odd n, $v_3(2^n + 1) = 1 + v_3(n)$. For $3^n - 1$: if n is odd, $v_2 = v_2(3 - 1) = 1$; if n is even, $v_2 = 1 + 2 + v_2(n) - 1 = 2 + v_2(n)$. □

7.3 Legendre's formula and Kummer's theorem

Legendre's formula

$$v_p(n!) = \sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - s_p(n)}{p - 1},$$

where $s_p(n)$ is the sum of the base-p digits of n. In particular $v_p(n!) < \frac{n}{p-1}$ and $v_2(n!) \leq n - 1$.

Proof. Among $1, \ldots, n$ there are $\lfloor n/p \rfloor$ multiples of p, $\lfloor n/p^2 \rfloor$ multiples of p^2, and so on; a number with $v_p = e$ is counted exactly e times. For the digit form, if $n = \sum_j n_j p^j$ then $\lfloor n/p^i \rfloor = \sum_{j \geq i} n_j p^{j-i}$, and summing over $i \geq 1$ gives $\sum_j n_j \frac{p^j - 1}{p-1}$. □

Kummer's theorem

$v_p\binom{m+n}{m}$ equals the number of carries when m and n are added in base p.

Proof. By Legendre, $v_p\binom{m+n}{m} = \frac{s_p(m) + s_p(n) - s_p(m+n)}{p-1}$. Each carry replaces p units in one position by one unit in the next, lowering the digit sum by $p - 1$. □

Use it when. Factorials or binomial coefficients appear in a divisibility. To prove $A \mid B$, show $v_p(A) \leq v_p(B)$ for each p; with Legendre this becomes an inequality between floor functions (Exercise 5). To prove an equation with a factorial has few solutions, compare the valuation of one prime on both sides (Problem 13).

7.4 Central binomial coefficients and the distribution of primes

Let $N = \binom{2n}{n}$. Three elementary facts about N control how primes are distributed:

1. $\frac{4^n}{2n+1} \le N \le 4^n$, since N is the largest of the $2n+1$ terms of $(1+1)^{2n}$.
2. Every prime p with $n < p \le 2n$ divides N (it divides $(2n)!$ but not $n!^2$).
3. If $p^a \mid N$ then $p^a \le 2n$: by Legendre $v_p(N) = \sum_i \left(\lfloor 2n/p^i \rfloor - 2\lfloor n/p^i \rfloor\right)$, each term is 0 or 1, and terms with $p^i > 2n$ vanish.

Facts 1 and 2 bound the product of the primes in $(n, 2n]$ by 4^n; Facts 1 and 3 show N needs many prime factors. Together they give estimates of the form $c_1 \frac{n}{\log n} < \pi(n) < c_2 \frac{n}{\log n}$ for the number $\pi(n)$ of primes up to n, and, with more care:

Bertrand's postulate (stated; proof omitted)

For every integer $n \ge 1$ there is a prime p with $n < p \le 2n$.

Example 7.3. *For $n \ge 2$, $n!$ is not a perfect square.*

Proof. Let $m = \lfloor n/2 \rfloor \ge 1$ and take a prime p with $m < p \le 2m \le n$. Then $2p > n$, so p is the only multiple of p up to n and $v_p(n!) = 1$, which is odd. □

Use it when. You need a prime in a prescribed range, usually one that divides a product exactly once.

8 Congruences for composite moduli (Chapter 7)

8.1 The Chinese remainder theorem

Chinese remainder theorem (CRT)

If $m_1, \dots, m_k$ are pairwise coprime, then for any integers $a_1, \dots, a_k$ the system $x \equiv a_i \pmod{m_i}$ $(i = 1, \dots, k)$ has a solution, and it is unique modulo $M = m_1 \cdots m_k$.

What it really says. Knowing x modulo M is the same as knowing x modulo each m_i, and the pieces can be prescribed *independently*. So you may design an integer one prime power at a time, and a congruence modulo n may be studied separately modulo each prime power of n.

Proof. Let $M_i = M/m_i$; it is coprime to m_i, so it has an inverse N_i modulo m_i. Then $x = \sum a_i M_i N_i$ works: modulo m_i all terms vanish except the i-th, which is a_i. If x, y are two solutions, each m_i divides $x - y$, hence so does M (pairwise coprime divisors). □

Use it when. (i) An existence problem with several independent divisibility demands ("n consecutive integers each divisible by ..."): assign one prime to each demand. (ii) Counting solutions of $f(x) \equiv 0 \pmod{n}$: multiply the counts for each prime power. (iii) Proving φ is multiplicative: x is coprime to mn iff it is coprime to m and to n.

8.2 Euler's theorem

Euler's theorem

If $\gcd(a, n) = 1$ then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Proof. Exactly as for Fermat: multiplication by a permutes the $\varphi(n)$ residues coprime to n; multiply them all and cancel. □

What it really says. Fermat's theorem for any modulus: exponents may be reduced modulo $\varphi(n)$. Typical use: last digits of towers of powers; existence of some k with $n \mid a^k - 1$ whenever $\gcd(a, n) = 1$.

8.3 Order

For $\gcd(a, n) = 1$, the *order* $\operatorname{ord}_n(a)$ is the smallest $d \geq 1$ with $a^d \equiv 1 \pmod{n}$.

The order lemma

$a^k \equiv 1 \pmod{n}$ if and only if $\operatorname{ord}_n(a) \mid k$. Consequently:

1. $\operatorname{ord}_n(a) \mid \varphi(n)$; for a prime p, $\operatorname{ord}_p(a) \mid p - 1$.
2. $a^i \equiv a^j \pmod{n}$ iff $i \equiv j \pmod{\operatorname{ord}_n(a)}$.
3. If p is prime and $p \mid a^k - 1$, then $\operatorname{ord}_p(a) \mid \gcd(k, p - 1)$.

Proof. Let $d = \operatorname{ord}_n(a)$ and $k = qd + r$, $0 \leq r < d$. Then $a^k \equiv a^r$, and $a^r \equiv 1$ with $r < d$ forces $r = 0$. The consequences follow from Euler and Fermat. □

What it really says. Fermat gives a period $p - 1$; the order is *the* period. Every exponent that works is a multiple of it. This converts information about exponents ($a^k \equiv 1$) into divisibility among integers ($d \mid k$ and $d \mid p - 1$), and so links the exponent to the prime.

Example 8.1. *Prove that if $n \geq 1$ and $n \mid 2^n - 1$, then $n = 1$.*

Proof. Suppose $n > 1$ and let p be its smallest prime divisor; p is odd. Since $p \mid 2^n - 1$, the order $d = \operatorname{ord}_p(2)$ divides $\gcd(n, p - 1)$. Every prime factor of $p - 1$ is less than p, so it does not divide n; thus $\gcd(n, p - 1) = 1$, $d = 1$, and $p \mid 2^1 - 1 = 1$, a contradiction. □

Example 8.2. *Let q be a prime. Every prime divisor p of $2^q - 1$ satisfies $p \equiv 1 \pmod{q}$; every prime divisor of $2^{2^k} + 1$ is $\equiv 1 \pmod{2^{k+1}}$.*

Proof. In the first case $\operatorname{ord}_p(2)$ divides q and is not 1, so it equals q, and $q \mid p - 1$. In the second, $2^{2^k} \equiv -1$ and $2^{2^{k+1}} \equiv 1$, so the order divides 2^{k+1} but not 2^k; it equals 2^{k+1} and divides $p - 1$. □

The wrong version. From $p \mid a^n - 1$ you cannot conclude $n \mid p - 1$. The correct statement is $\operatorname{ord}_p(a) \mid n$ *and* $\operatorname{ord}_p(a) \mid p - 1$.

8.4 Primitive roots

Primitive roots modulo a prime

For every prime p there is g with $\operatorname{ord}_p(g) = p-1$. Then $1, g, g^2, \ldots, g^{p-2}$ are all the non-zero residues modulo p. (More generally, primitive roots exist exactly for the moduli $1, 2, 4, p^k, 2p^k$ with p odd.)

What it really says. The non-zero residues modulo p form one cycle under multiplication. Writing $x \equiv g^i$ turns multiplication modulo p into addition of exponents modulo $p-1$, a discrete logarithm. Statements about powers become linear congruences in the exponents.

Proof. Let $\psi(d)$ be the number of residues of order d, for $d \mid p-1$. If some a has order d, then $1, a, \ldots, a^{d-1}$ are d distinct roots of $x^d - 1$, hence all of them (Lagrange). So every element of order d is some a^k, and a^k has order d exactly when $\gcd(k, d) = 1$. Thus $\psi(d)$ is 0 or $\varphi(d)$. But $\sum_{d \mid p-1} \psi(d) = p - 1 = \sum_{d \mid p-1} \varphi(d)$, so $\psi(d) = \varphi(d)$ for all d, and $\psi(p-1) = \varphi(p-1) > 0$. □

Consequences worth remembering.

- $x^k \equiv a \pmod p$ with $p \nmid a$ is solvable iff $a^{(p-1)/\gcd(k,p-1)} \equiv 1$. There are $\frac{p-1}{\gcd(k,p-1)}$ non-zero k-th powers. If $\gcd(k, p-1) = 1$, every residue is a k-th power.
- $1^k + 2^k + \cdots + (p-1)^k \equiv -1 \pmod p$ if $p - 1 \mid k$, and $\equiv 0$ otherwise (a geometric series in g^k).

Use it when. The problem concerns k-th powers modulo p, sums of powers over all residues, or you need an element of a prescribed order.

9 Competition problems: from signal to solution

All thirteen problems are from the International Mathematical Olympiad. Each has a box that records the reasoning *before* the solution: the signal in the statement, the tool it suggests, and the plan. The line after each solution ties the problem back to the maps.

Problem 1 (IMO 1959, Problem 1). *Prove that the fraction* $\dfrac{21n+4}{14n+3}$ *is irreducible for every natural number* n.

How to think. **Signal:** "irreducible" means gcd = 1, and both expressions are linear in n. **Tool:** a common divisor divides every integer combination. **Plan:** find a combination in which n cancels.

Solution. $3(14n+3) - 2(21n+4) = 1$. Any common divisor of numerator and denominator divides 1. □

Link. Map 2, row 3 (Euclid-style reduction); Bézout's certificate $au + bv = 1$.

Problem 2 (IMO 1964, Problem 1). *(a) Find all positive integers* n *for which* $2^n - 1$ *is divisible by* 7. *(b) Prove that there is no positive integer* n *for which* $2^n + 1$ *is divisible by* 7.

How to think. **Signal:** a fixed base, a varying exponent, a fixed modulus. **Tool:** powers are periodic; find the order. **Plan:** compute $\operatorname{ord}_7(2)$ and list one period.

Solution. $2^1, 2^2, 2^3 \equiv 2, 4, 1 \pmod 7$, so $\operatorname{ord}_7(2) = 3$. (a) By the order lemma $7 \mid 2^n - 1$ iff $3 \mid n$. (b) 2^n takes only the values $1, 2, 4$ modulo 7, so $2^n + 1 \in \{2, 3, 5\}$, never 0. □

Link. Order lemma. Part (b) returns as the last step of Problem 7.

Problem 3 (IMO 1979, Problem 1). *Let p and q be natural numbers such that* $\frac{p}{q} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{1318} + \frac{1}{1319}$. *Prove that p is divisible by* 1979.

> **How to think.** **Signal:** a sum of reciprocals and a prime; note $1979 = 660 + 1319$. **Tool:** fractions modulo a prime; pair k with $1979 - k$. **Plan:** rewrite the alternating sum as a plain sum over a symmetric range, then pair terms to pull out the factor 1979.

Solution. Adding and subtracting the even terms,

$$\frac{p}{q} = \sum_{k=1}^{1319} \frac{1}{k} - 2\sum_{k=1}^{659} \frac{1}{2k} = \sum_{k=660}^{1319} \frac{1}{k}.$$

The range $660, \ldots, 1319$ is symmetric about $1979/2$, so pairing k with $1979 - k$,

$$\frac{p}{q} = \sum_{k=660}^{989} \left(\frac{1}{k} + \frac{1}{1979 - k}\right) = 1979 \sum_{k=660}^{989} \frac{1}{k(1979 - k)} = 1979 \cdot \frac{A}{B},$$

where B is a product of integers smaller than 1979. Since 1979 is prime, $\gcd(B, 1979) = 1$. From $pB = 1979Aq$ and Gauss' lemma, $1979 \mid p$. □

Link. Fractions modulo p; the pairing in the proof of Wolstenholme's theorem.

Problem 4 (IMO 1986, Problem 1). *Let d be a positive integer different from* 2, 5, 13. *Show that one can find distinct a, b in the set $\{2, 5, 13, d\}$ such that $ab - 1$ is not a perfect square.*

> **How to think.** **Signal:** "is not a perfect square". **Tool:** squares take few values modulo 4, 8, 16. **Plan:** $2 \cdot 5 - 1$, $2 \cdot 13 - 1$, $5 \cdot 13 - 1$ are squares, so assume $2d - 1$, $5d - 1$, $13d - 1$ are all squares and seek a parity contradiction. Use the first to learn the parity of d, then combine the other two.

Solution. Suppose $2d - 1 = x^2$, $5d - 1 = y^2$, $13d - 1 = z^2$. Then x is odd, $x = 2u + 1$, so $d = 2u^2 + 2u + 1$ is odd. Hence y and z are even, $y = 2y_1$, $z = 2z_1$, and subtracting, $z^2 - y^2 = 8d$ gives $z_1^2 - y_1^2 = 2d$. But $2d \equiv 2 \pmod 4$, while a difference of two squares is 0, 1 or 3 modulo 4. Contradiction. □

Link. Map 2, row 1 (choose a modulus). Combining two conditions by subtraction removed the unknown constant.

Problem 5 (IMO 1988, Problem 6). *Let a and b be positive integers such that $ab + 1$ divides $a^2 + b^2$. Show that* $\frac{a^2 + b^2}{ab + 1}$ *is the square of an integer.*

> **How to think.** **Signal:** a symmetric condition, quadratic in each variable, with a conclusion about the *value* of the quotient. **Tool:** Vieta jumping. **Plan:** fix the quotient k, view the relation as a quadratic in a, and replace a by the other root. Ask what stops the descent: it

can only stop at a root equal to 0, and that forces $k = b^2$.

Solution. Let $k = \frac{a^2+b^2}{ab+1}$ and suppose k is not a perfect square. Among all pairs of positive integers (a, b) giving this k, choose one with $a + b$ minimal, and let $a \geq b$. The number a is a root of

$$x^2 - kb\,x + (b^2 - k) = 0.$$

Let a' be the other root. By Vieta, $a' = kb - a$, so a' is an integer, and $aa' = b^2 - k$.

$a' \neq 0$: otherwise $k = b^2$ is a square. $a' > 0$: if $a' < 0$ then $a'^2 - kba' + b^2 - k \geq a'^2 + k + b^2 - k > 0$, contradicting that a' is a root. $a' < a$: $a' = \frac{b^2-k}{a} < \frac{b^2}{a} \leq a$.

So (a', b) is a pair of positive integers with the same k and $a' + b < a + b$, contradicting minimality. Hence k is a perfect square. □

Link. Map 2, row 6. The three checks (integer, positive, smaller) are exactly the descent checklist of Section 4.5.

Problem 6 (IMO 1989, Problem 5). *Prove that for each positive integer n there exist n consecutive positive integers none of which is an integral power of a prime number.*

How to think. **Signal:** "there exist n consecutive integers each with a property"; the property can be forced by divisibility. **Tool:** CRT, one condition per integer. **Plan:** a number divisible by two different primes is not a prime power; so demand that $x + i$ be divisible by two primes reserved for i.

Solution. Choose $2n$ distinct primes $p_1, \ldots, p_n, q_1, \ldots, q_n$. The moduli $p_i q_i$ are pairwise coprime, so by CRT there is a positive integer x with $x \equiv -i \pmod{p_i q_i}$ for $i = 1, \ldots, n$. Then $x + i$ is divisible by the two distinct primes p_i, q_i and is not a power of a prime. □

Link. Map 2, row 7 (build n prime by prime).

Problem 7 (IMO 1990, Problem 3). *Determine all integers $n > 1$ such that $\dfrac{2^n + 1}{n^2}$ is an integer.*

How to think. **Signal:** n appears both as exponent and as divisor. **Tools:** smallest prime divisor with order, to find which prime divides n; then LTE, because n^2 asks for a high power of that prime while $v_p(2^n + 1)$ grows slowly. **Plan:** (1) smallest prime is 3; (2) LTE limits the power of 3 to one; (3) repeat the smallest-prime argument on the rest.

Solution. n is odd. *Step 1.* Let p be the smallest prime divisor of n. From $2^n \equiv -1$ we get $2^{2n} \equiv 1 \pmod{p}$, so $\operatorname{ord}_p(2) \mid \gcd(2n, p-1) = 2$ (as $\gcd(n, p-1) = 1$). Thus $p \mid 2^2 - 1$, $p = 3$.

Step 2. Write $n = 3^k m$ with $3 \nmid m$, $k \geq 1$. By LTE (n odd, $3 \mid 2 + 1$), $v_3(2^n + 1) = 1 + k$. We need $2k = v_3(n^2) \leq k + 1$, so $k = 1$.

Step 3. Suppose $m > 1$ and let $q \geq 5$ be its smallest prime divisor. As before $\operatorname{ord}_q(2) \mid \gcd(6m, q-1)$, which divides 6 because $\gcd(m, q-1) = 1$. So $q \mid 2^6 - 1 = 63$, giving $q = 7$. But $2^n + 1$ is never divisible by 7 (Problem 2b). Hence $m = 1$.

So $n = 3$, and indeed $2^3 + 1 = 9 = 3^2$. □

Link. Map 2, rows 4 and 5 in succession. Note that Step 3 re-argues for the next prime instead of assuming it behaves like the smallest.

Problem 8 (IMO 1999, Problem 4). *Determine all pairs (n, p) of positive integers such that p is a prime, $n \le 2p$, and $(p-1)^n + 1$ is divisible by n^{p-1}.*

> **How to think.** **Signal:** the same shape as Problem 7, with base $p-1$. **Tools:** smallest prime divisor and order, then a valuation count. **Plan:** show the smallest prime of n is p itself; the bound $n \le 2p$ then forces $n = p$; finally count $v_p((p-1)^p + 1)$.

Solution. $n = 1$ works for every p. If $p = 2$, the condition is $n \mid 2$ with $n \le 4$: $n = 1, 2$. Now let $p \ge 3$ and $n > 1$. Then $(p-1)^n + 1$ is odd, so n is odd, and $n < 2p$.

Let q be the smallest prime divisor of n. Since $(p-1)^{2n} \equiv 1 \pmod q$, the order of $p-1$ modulo q divides $\gcd(2n, q-1) = 2$. So $q \mid (p-1)^2 - 1 = p(p-2)$. If $q \mid p-2$ then $p-1 \equiv 1$ and $(p-1)^n + 1 \equiv 2 \pmod q$, impossible for odd q. Hence $q = p$, so $p \mid n$, and $n < 2p$ gives $n = p$.

Now we need $p^{p-1} \mid (p-1)^p + 1$. By LTE, $v_p((p-1)^p + 1^p) = v_p(p) + v_p(p) = 2$. So $p - 1 \le 2$, $p = 3$, and $2^3 + 1 = 9 = 3^2$ works.

Answer: $(1, p)$ for every prime p, $(2, 2)$ and $(3, 3)$. □

Link. Same route as Problem 7. Recognising a problem as a known shape is half the solution.

Problem 9 (IMO 2005, Problem 4). *Consider the sequence $a_n = 2^n + 3^n + 6^n - 1$ $(n = 1, 2, \dots)$. Determine all positive integers that are relatively prime to every term of the sequence.*

> **How to think.** **Signal:** we must decide which primes divide *some* term; the index is ours to choose. **Tool:** Fermat: choose n in terms of p so that the powers collapse. **Plan:** $n = p - 1$ gives $1 + 1 + 1 - 1 = 2$, no use. Try $n = p - 2$, where $a^{p-2} \equiv a^{-1}$: the sum becomes $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} - 1 = 0$.

Solution. The answer is only 1; we show every prime divides some term. First, $a_2 = 48$ is divisible by 2 and 3. For a prime $p \ge 5$, by Fermat,

$$6a_{p-2} = 3 \cdot 2^{p-1} + 2 \cdot 3^{p-1} + 6^{p-1} - 6 \equiv 3 + 2 + 1 - 6 = 0 \pmod p,$$

and $p \nmid 6$, so $p \mid a_{p-2}$. An integer greater than 1 has a prime factor, which divides some term. □

Link. Fermat's theorem as a way to *manufacture* divisibility. The identity $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$ is the hidden reason for the numbers $2, 3, 6$.

Problem 10 (IMO 1998, Problem 4). *Determine all pairs (a, b) of positive integers such that $ab^2 + b + 7$ divides $a^2b + a + b$.*

> **How to think.** **Signal:** one polynomial divides another of similar size. **Tools:** linear combination to lower the degree, then divisibility ⇒ inequality. **Plan:** eliminate the leading terms; the divisor must then divide something small, which is either 0 or at least as large as the divisor in absolute value.

Solution. Let $D = ab^2 + b + 7$. Then D divides

$$b(a^2b + a + b) - a(ab^2 + b + 7) = b^2 - 7a.$$

Case $b^2 - 7a > 0$. Then $0 < b^2 - 7a < b^2 < D$, impossible.

Case $b^2 - 7a = 0$. Then $7 \mid b$; put $b = 7k$, $a = 7k^2$. Check: $D = 7(49k^4 + k + 1)$ and $a^2b + a + b = 7k(49k^4 + k + 1)$, so every such pair works.

Case $b^2 - 7a < 0$. Then $D \le 7a - b^2 < 7a$, so $ab^2 < 7a$ and $b \in \{1, 2\}$. If $b = 1$: $a + 8 \mid 7a - 1 = 7(a + 8) - 57$, so $a + 8 \mid 57$, giving $a = 11$ or $a = 49$; both work ($133 = 7 \cdot 19$ and $2451 = 43 \cdot 57$). If $b = 2$: $4a + 9 \mid 7a - 4$, hence $4a + 9 \mid 7(4a + 9) - 4(7a - 4) = 79$, so $4a + 9 = 79$, which has no integer solution.

Answer: $(11, 1)$, $(49, 1)$ and $(7k^2, 7k)$ for $k \ge 1$. □

Link. Map 2, rows 2 and 3. The conditions derived were only necessary, so each candidate was checked in the original statement.

Problem 11 (IMO 2006, Problem 4). *Determine all pairs* (x, y) *of integers such that* $1 + 2^x + 2^{2x+1} = y^2$.

> **How to think.** **Signal:** an exponential equation with a perfect square; moving 1 across gives a difference of squares. **Tools:** factor; the two factors $y \pm 1$ have gcd 2, so almost all powers of 2 sit in one of them; then compare sizes. **Plan:** write $y = 2^{x-1}m \pm 1$ and substitute.

Solution. If (x, y) is a solution so is $(x, -y)$; take $y > 0$. For $x \le -2$ the left side is not an integer, and $x = -1$ gives 2. Direct checking: $x = 0$ gives $4 = 2^2$; $x = 1, 2$ give 11 and 37. Let $x \ge 3$. Then y is odd and

$$2^x \left(1 + 2^{x+1}\right) = (y - 1)(y + 1).$$

One of $y \pm 1$ is $\equiv 2 \pmod 4$, so the other is divisible by 2^{x-1}: $y = 2^{x-1}m + \varepsilon$ with $m \ge 1$, $\varepsilon = \pm 1$. Substituting, $y^2 - 1 = 2^{2x-2}m^2 + 2^x m\varepsilon$, and dividing by 2^x,

$$1 - \varepsilon m = 2^{x-2}\left(m^2 - 8\right).$$

If $\varepsilon = 1$*:* for $m = 1, 2$ the left side is 0 or -1, while the right side is $-7 \cdot 2^{x-2}$ or $-4 \cdot 2^{x-2}$, which is at most -8; for $m \ge 3$ the left side is negative and the right side positive. No solutions.

If $\varepsilon = -1$*:* $1 + m = 2^{x-2}(m^2 - 8) \ge 2(m^2 - 8)$ requires $m^2 > 8$ and $2m^2 - m - 17 \le 0$, so $m = 3$. Then $4 = 2^{x-2}$, $x = 4$, $y = 23$, and indeed $1 + 16 + 512 = 529 = 23^2$.

Answer: $(0, \pm 2)$ and $(4, \pm 23)$. □

Link. Valuation of a single prime (v_2 of the two factors) followed by a size comparison: two measurements in succession.

Problem 12 (IMO 2003, Problem 6). *Let* p *be a prime number. Prove that there exists a prime number* q *such that for every integer* n*, the number* $n^p - p$ *is not divisible by* q.

> **How to think.** **Signal:** we want p *not* to be a p-th power modulo q. **Tool:** order. If $n^p \equiv p$, then the order of n is tied to the order of p. **Plan:** pick q so that p has order exactly p modulo q (so $q \mid \frac{p^p - 1}{p - 1}$) and so that $p^2 \nmid q - 1$; then the order of n is squeezed to divide p, forcing $p \equiv 1$, a contradiction.

Solution. Let $N = \frac{p^p - 1}{p - 1} = 1 + p + \cdots + p^{p-1}$. Then $N \equiv 1 + p \pmod{p^2}$. If every prime factor of N were $\equiv 1 \pmod{p^2}$, so would N be. So N has a prime factor q with $q \not\equiv 1 \pmod{p^2}$. Note $q \ne p$ since $N \equiv 1 \pmod p$.

Suppose $n^p \equiv p \pmod q$ for some n. Then $q \nmid n$ and $n^{p^2} \equiv p^p \equiv 1 \pmod q$. So $\operatorname{ord}_q(n)$ divides $\gcd(p^2, q-1)$, which is 1 or p because $p^2 \nmid q-1$. Hence $n^p \equiv 1$, that is, $p \equiv 1 \pmod q$. But then $N = 1 + p + \cdots + p^{p-1} \equiv p \pmod q$, so $q \mid p$ and $q = p$, a contradiction. □

Link. Order lemma twice, with Euclid's idea of choosing a prime factor of a constructed number whose residue class is forced.

Problem 13 (IMO 2019, Problem 4). *Find all pairs (k, n) of positive integers such that*

$$k! = (2^n - 1)(2^n - 2)(2^n - 4) \cdots (2^n - 2^{n-1}).$$

How to think. **Signal:** a factorial equals a product whose power of 2 is easy to read. **Tools:** Legendre's bound $v_2(k!) < k$ gives a *lower* bound on k; the size of the right side gives an *upper* bound. **Plan:** show the two bounds clash for large n and check small n by hand.

Solution. The factor $2^n - 2^i = 2^i(2^{n-i} - 1)$ has $v_2 = i$, so the right side R has $v_2(R) = 0+1+\cdots+(n-1) = \frac{n(n-1)}{2}$. By Legendre, $v_2(k!) \le k-1$. Hence $k > \frac{n(n-1)}{2}$. Also $R < (2^n)^n = 2^{n^2}$.

Claim: $\left(\frac{n(n-1)}{2}\right)! > 2^{n^2}$ *for* $n \ge 6$. For $n = 6$: $15! > 1.3 \cdot 10^{12} > 2^{36}$. Passing from n to $n+1$ multiplies the left side by n integers each larger than $\frac{n(n-1)}{2} \ge 15$, hence by at least $16^n = 2^{4n}$, and the right side by 2^{2n+1}.

So for $n \ge 6$, $k! > \left(\frac{n(n-1)}{2}\right)! > 2^{n^2} > R$: no solutions. For $n = 1, \ldots, 5$, the values of R are 1, 6, 168, 20160 and 9999360. Only $1 = 1!$ and $6 = 3!$ are factorials, since $5! < 168 < 6!$, $7! < 20160 < 8!$ and $10! < 9999360 < 11!$.

Answer: $(k, n) = (1, 1)$ and $(3, 2)$. □

Link. Map 2, row 8 with row 2: a valuation gives one inequality, size gives the opposite one.

10 Recognition table

Feature of the problem	**First moves, in order**
"No integer solutions"; "is not a perfect square/cube"	Residues mod $3, 4, 8, 16$ (squares), $7, 9$ (cubes); then a prime p with $k \mid p-1$ for k-th powers.
$A \mid B$ with A, B polynomials of similar size	Combine to lower the degree; then $\lvert A\rvert \le \lvert$remainder$\rvert$ or remainder $= 0$; check small cases.
gcd of two expressions; "coprime"; "irreducible"	Euclid's step $\gcd(A, B) = \gcd(A - kB, B)$ until a constant.
$n \mid a^n \pm 1$ or exponent and modulus linked	Smallest prime p of n; $\operatorname{ord}_p(a) \mid \gcd(\cdot, p-1)$; then LTE for the power of p.
Exact power of p in $a^n \pm b^n$; $a^x - b^y = c$	LTE (check hypotheses); compare with the size of the other side.
Product = perfect power	Show factors are coprime (or have small gcd); each is a perfect power up to that gcd.
Symmetric quadratic relation in two variables	Vieta jumping on a minimal solution; check integer, positive, smaller.
"Only the trivial solution", homogeneous equation	Infinite descent via divisibility by a fixed prime.
"Prove there exist . . . " with several divisibility demands	CRT with one prime (or prime power) per demand; Hensel if prime powers are needed.
Factorials, binomial coefficients, "is an integer"	v_p on both sides: Legendre, Kummer, Lucas (mod p only).
$p \mid x^2 + 1$, $p \mid x^2 + ay^2$, sums of two squares	-1 is a square iff $p \equiv 1 \pmod 4$; Legendre symbol; reciprocity.
"Infinitely many primes of the form . . . "	Euclid: construct N coprime to known primes whose prime factors are forced into the class.
Sum of reciprocals modulo p	Pair k with $p - k$; replace k^{-1} by a permutation of residues.
"Some sum/difference is divisible by n"	Pigeonhole on $n + 1$ partial sums or values.
"Infinitely many solutions" of a quadratic equation	Reduce to Pell $x^2 - dy^2 = 1$; generate from one solution.

11 Common mistakes

1. Cancelling in a congruence. $ac \equiv bc \pmod n$ gives $a \equiv b$ only modulo $n/\gcd(c, n)$. Cancel freely only when $\gcd(c, n) = 1$.

2. Fermat with $p \mid a$, or reducing exponents modulo p. $a^{p-1} \equiv 1$ requires $p \nmid a$. Exponents reduce modulo $p - 1$ (or modulo the order), never modulo p.

3. "$p \mid a^n - 1$, so $n \mid p - 1$." False: $7 \mid 2^6 - 1$ and $6 \mid 6$, but also $7 \mid 2^9 - 1$ while $9 \nmid 6$. Correct: $\operatorname{ord}_p(a)$ divides both n and $p - 1$.

4. LTE without its hypotheses. $v_3(5^n - 1)$ is *not* $v_3(4) + v_3(n)$, because $3 \nmid 5 - 1$. Here $\operatorname{ord}_3(5) = 2$: for odd n the valuation is 0; for even $n = 2m$ apply LTE to $25^m - 1$.

5. Lagrange for composite moduli. A polynomial of degree d may have more than d roots modulo a composite number. Use CRT to split the modulus first.

6. Necessary conditions presented as answers. After deriving candidates from consequences of the condition, substitute each one back. In "find all" problems the verification is part of the solution.

7. A descent that does not descend. The new solution must be shown to be an integer, admissible, and strictly smaller in a quantity bounded below.

12 Practice problems

Method labels are deliberately absent; choosing the tool is part of the exercise. Full solutions follow.

1. Find all positive integers n such that $n + 1$ divides $n^2 + 1$.
2. Prove that $x^2 + y^2 + z^2 = 2xyz$ has no solutions in positive integers.
3. Find all positive integers n such that 2^n divides $3^n - 1$.
4. Prove that for every n there are n consecutive positive integers each divisible by a perfect square greater than 1.
5. (IMO 1972, Problem 3) Let m and n be non-negative integers. Prove that $\dfrac{(2m)!\,(2n)!}{m!\,n!\,(m+n)!}$ is an integer.
6. Prove that no prime of the form $8k + 7$ divides a number of the form $2^n + 1$.

13 Solutions to the practice problems

1. $n^2 + 1 = (n + 1)(n - 1) + 2$, so $n + 1 \mid 2$ and $n = 1$. *Idea:* reduce, then divisor $\leq$ remainder.

2. We prove more: for every $k \geq 1$, $x^2 + y^2 + z^2 = 2^k xyz$ has no positive solutions. Suppose it has, for some k, and take a solution with $x + y + z$ minimal over all $k \geq 1$. The right side is even, so either all of x, y, z are even or exactly one is. If exactly one is even, the left side is $\equiv 2 \pmod 4$ while the right side is divisible by $2^k \cdot 2$, hence by 4. So all are even; writing $x = 2x_1$, etc., gives $x_1^2 + y_1^2 + z_1^2 = 2^{k+1} x_1 y_1 z_1$, a smaller solution of the same family. Contradiction. *Idea:* descent; the family had to be enlarged so that the new equation is again of the assumed kind.

3. If n is odd, $v_2(3^n - 1) = 1$, so only $n = 1$. If n is even, LTE gives $v_2(3^n - 1) = 2 + v_2(n)$. With $v = v_2(n)$ we need $2^v \leq n \leq 2 + v$, so $v \leq 2$ and $n \leq 4$: $n = 2$ ($v_2 = 3 \geq 2$) and $n = 4$ ($v_2 = 4 \geq 4$) work. *Answer:* $n = 1, 2, 4$. *Idea:* valuation grows like $\log n$, the demand grows like n.

4. Take distinct primes $p_1, \ldots, p_n$ and, by CRT, a positive x with $x \equiv -i \pmod{p_i^2}$. Then $p_i^2 \mid x + i$. *Idea:* one prime per demand.

5. By Legendre it suffices that for every prime power $q = p^i$,

$$\left\lfloor \frac{2m}{q} \right\rfloor + \left\lfloor \frac{2n}{q} \right\rfloor \geq \left\lfloor \frac{m}{q} \right\rfloor + \left\lfloor \frac{n}{q} \right\rfloor + \left\lfloor \frac{m+n}{q} \right\rfloor.$$

This is $\lfloor 2\alpha \rfloor + \lfloor 2\beta \rfloor \geq \lfloor \alpha \rfloor + \lfloor \beta \rfloor + \lfloor \alpha + \beta \rfloor$ for real $\alpha, \beta \geq 0$. Both sides change by the same amount when α or β increases by 1, so assume $0 \leq \alpha, \beta < 1$. The right side is then $\lfloor \alpha + \beta \rfloor \in \{0, 1\}$, and if it is 1 then one of α, β is at least $\frac{1}{2}$, making the left side at least 1. *Idea:* local–global; an integrality claim becomes a floor inequality.

6. Let $p \mid 2^n + 1$, p odd. If n is even, $-1 \equiv (2^{n/2})^2$ is a square, so $p \equiv 1 \pmod 4$. If n is odd, $-2 \equiv (2^{(n+1)/2})^2$ is a square, so $\left(\frac{-1}{p}\right)\left(\frac{2}{p}\right) = 1$; checking the four odd classes modulo 8, this holds only for $p \equiv 1, 3 \pmod 8$. In neither case is $p \equiv 7 \pmod 8$. *Idea:* turn "p divides" into "something is a square modulo p".

14 Compact revision sheet

Four moves. Reduce modulo something • compare sizes • compare prime powers • use the cycle of powers.

Tool	Statement with hypotheses
Size	$a \mid b$, $b \neq 0 \Rightarrow \lvert a\rvert \leq \lvert b\rvert$.
Bézout	$\{ax+by\} = \gcd(a,b)\mathbb{Z}$; a invertible mod n iff $\gcd(a,n)=1$.
Gauss	$a \mid bc$, $\gcd(a,b)=1 \Rightarrow a \mid c$.
Coprime factors	$\gcd(a,b)=1$, $ab=c^k \Rightarrow a,b$ are k-th powers $(a,b>0)$.
Fermat / Euler	$p \nmid a \Rightarrow a^{p-1} \equiv 1\ (p)$; $\gcd(a,n)=1 \Rightarrow a^{\varphi(n)} \equiv 1\ (n)$.
Wilson	$(p-1)! \equiv -1\ (p)$; characterises primes.
Lagrange	Degree $d \Rightarrow$ at most d roots mod a *prime*.
-1 a square	$x^2 \equiv -1\ (p)$ solvable iff $p=2$ or $p \equiv 1\ (4)$.
Euler's criterion	$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2}\ (p)$; $\left(\frac{2}{p}\right)=1$ iff $p \equiv \pm 1\ (8)$.
Order	$a^k \equiv 1\ (n) \iff \operatorname{ord}_n(a) \mid k$; so $\operatorname{ord}_p(a) \mid \gcd(k, p-1)$.
Primitive root	Some g has order $p-1$ mod p.
CRT	Pairwise coprime moduli: residues can be prescribed independently.
v_p	$v_p(ab)=v_p(a)+v_p(b)$; $v_p(a+b) \geq \min$, equality if different.
LTE	p odd, $p \mid a-b$, $p \nmid ab$: $v_p(a^n-b^n)=v_p(a-b)+v_p(n)$.
LTE, $p=2$	a,b odd, n even: $v_2(a^n-b^n)=v_2(a-b)+v_2(a+b)+v_2(n)-1$.
Legendre	$v_p(n!)=\sum\lfloor n/p^i\rfloor = \frac{n-s_p(n)}{p-1}$.
Kummer / Lucas	$v_p\binom{m+n}{m}$ = carries in base p; $\binom{n}{k} \equiv \prod \binom{n_i}{k_i}\ (p)$.
Hensel	$f(a) \equiv 0$, $f'(a) \not\equiv 0\ (p) \Rightarrow$ unique lift mod p^k.
Bertrand	A prime in $(n, 2n]$ for every $n \geq 1$.

Before writing up, check: hypotheses of each theorem quoted; exceptional primes (2, 3, and any p dividing a base); small cases; that every candidate answer satisfies the original condition.